

YATISHA RAJANALA

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PROFESSIONAL SUMMARY:

I am a resourceful Health Informatics professional with expertise in data analytics, epidemiology, and machine learning; dedicated to improving healthcare outcomes through advanced statistical and predictive modeling techniques. I am skilled in managing large healthcare datasets, integrating IT standards, and utilizing diverse programming tools; and aspire to develop actionable insights to optimize clinical workflows and patient care. As a proactive collaborator, I am committed to driving innovation in health data analytics and personalized healthcare interventions.

EDUCATION:

Master of Science in Health Informatics with Health Data Analytics Concentration

George Mason University, Fairfax, VA

January 2023 – December 2024

CGPA: 4.0

Doctor of Pharmacy (PharmD)

Jawaharlal Nehru Technological University, Hyderabad, India

August 2016 – August 2022

CGPA: 3.8

SKILLS:

Programming Languages:	R, Python, Linux, MATLAB, STATA
Programming Tools and Software:	Microsoft Office Suite, Tableau, SPSS, SAS, Netica, Lucidchart, Zotero
Web Technologies and Tools:	HTML, CSS, REGEX, POSTMAN
Databases and Data Analysis Tools:	SQL, WEKA, NDSR
Healthcare IT and Standards:	XML, HL7, SNOMED, ICD, CDA, FHIR, GEMS
Quantitative and Analytical Skills:	Machine Learning, Statistical Analysis, Data Mining, Data Modelling, Predictive Modelling, Artificial Intelligence, Natural Language Processing, Dashboards, and Reporting
Spoken Languages:	English, Telugu, Hindi, French

RELATED WORK EXPERIENCE:

Research Assistant

George Mason University, Fairfax, VA

September 2023 – Present

- Collaborated with Professors and Peers in the Departments of Nutrition & Food Studies and Health Administration & Policy at the College of Public Health.
- Utilized Citrix, Python and SQL to extract, consolidate, and clean the datasets pertaining to National Health and Nutrition Examination Survey (NHANES) to combine participant data from different years and categories of questionnaire, examination, laboratory, and dietary data.
- Analyzed the datasets using RStudio, SPSS, SAS, STATA, and Microsoft Tools to perform statistical analyses to determine the significant trends over the years for diabetic populations, with special focus on nutrition, macronutrients intake, depression, cardiovascular diseases, and sleep.
- Performed multinomial logistic regression to analyze the trends of depression in non-diabetes, prediabetes, and diabetes subpopulations, with special considerations given to BMI and race/ethnicity.
- Managed datasets from the Automated Self-Administered 24-Hour (ASA24) Dietary Assessment Tool, to assess the trends in sleep and nutrition using the Alternate Healthy Eating Index (AHEI) & Percentage of Micronutrients intake of magnesium, vitamin D, zinc and potassium, utilizing RStudio, STATA and Python.
- Developed a machine learning project utilizing the 30-day continuous dietary recall data of participants from the Individualized Nutrition Therapy (INT) study, to closely monitor the foods that cause a glucose spike, prolonged duration of spike (calculated using the Area Under Curve (AUC)), and estimate the macronutrients and total calories for each participant; to make the best possible nutritional intervention or a personalized dietary plan.
- Evaluated the accuracy and limitations of the AI-generated outputs by comparing them with manually analyzed 30-day dietary recall data, to assess the feasibility of developing an AI-based nutritional analysis model against traditional software tools like NDSR.
- Performed solid-phase sandwich Enzyme-Linked Immunosorbent Assay (ELISA) for the INT study for determining the presence of Endothelium-1 in the serum of the patients' control group, post consumption of mangoes for 30 days, to test whether mangoes can reduce the presence of cardiovascular inflammatory markers.
- Similar ELISA tests were conducted to measure C-Reactive Protein and Insulin levels in participants of the Individualized Nutrition Therapy (INT) study, aimed at evaluating changes in dietary and sleep patterns before and after the food interventions.

Healthcare Data Analyst Intern

Avalon In-Home Nursing and Rehabilitation, L.L.C., Falls Church, VA

September 2024 – December 2024

- Coordinated with the Director of Nursing, to gather the Outcome and Assessment Information Set (OASIS) datasets relevant to patients of Avalon In-Home Nursing and Rehabilitation, L.L.C.

- XML patient data from September 2023 to August 2024 was collected; and data privacy and analytical consistency was maintained by anonymizing the datasets using Python.
- Utilizing RStudio, the conversion of date of births to age, calculation of Body Mass Index (BMI), categorization of age and BMI into defined groups, and identification of the use of high-risk drugs was done.
- Correlation analysis using heatmaps, and logistic regression analysis with statistically significant confidence intervals and *p*-values, were done in RStudio, to identify the predominant clinical and functional predictors of acute care rehospitalizations.
- This project analysis identified factors such as exhaustion, polypharmacy, cognitive decline, and mobility limitations, to be of utmost importance while diagnosing and managing these patient subpopulations.
- The implementation of the findings has been found to improve 18.6% of the organizations' working environment, with clear demarcations of the workflow using Tableau.

VOLUNTEER EXPERIENCE:

Doctor of Pharmacy Internship

Mahavir Hospital, Vikarabad, Telangana, India

August 2021 – August 2022

- Apprenticeship in the Departments of General Medicine, Pediatrics, Dermatology, and Gynecology, with special attention to prescribing medications, drug monitoring, and patient counselling.

Pharmacist Internship

Apoorva Hospital, Hyderabad, Telangana, India

November 2021 – December 2021

- Assisted in the hospital pharmacy and collaborated with physicians in the Department of Pediatrics with specialized care for neonates, drug prescriptions, ward rounds, drug dosing, and health record management.

Clinical Pharmacologist Internship

Apollo Health City, Hyderabad, Telangana, India

August 2021 – November 2021

- Interned as a Clinical Pharmacologist in the Departments of Pulmonology, Internal Medicine, Oncology, Podiatry, Nephrology, Cardiology and Gastroenterology.

Physician Assistant Internship

Krishna Institute of Medical Sciences (KIMS), Hyderabad, Telangana, India

May 2018 – June 2018

- Summer Internship in the Department of Orthopedics, to closely monitor the adverse drug reactions, side effects of polypharmacy, and drug dispensary within the operating rooms.

PUBLICATIONS:

Basiri, R., & Rajanala, Y. (2025). Effects of Individualized Nutrition Therapy and Continuous Glucose Monitoring on Dietary and Sleep Quality in Individuals with Prediabetes and Overweight or Obesity. *Nutrients*, 17(9), 1507. <https://doi.org/10.3390/nu17091507>

Basiri, R., Rajanala, Y., Kassem, M., Cheskin, L. J., Frankenfeld, C. L., & Farvid, M. S. (2024). Diabetes Control Status and Severity of Depression: Insights from NHANES 2005–2020. *Biomedicines*, 12(10), Article 10. <https://doi.org/10.3390/biomedicines12102276>

Praveena Devi, CH. B., Yatisha, R., Sharma, JVC, Sirisha, P. & Sudha Rani, G. (2019). Methemoglobinemia: A Differential Diagnosis Not To Be Overlooked. *Adalya Journal*, Vol. 8 (9), ISSN No. 1301-2746. <https://www.dropbox.com/scl/fi/1062vzhng52hcff7e9thd/78-Sep-1741.pdf?dl=0&rlkey=qepfkamzotgv8xwqjrrs4v620>

Yatisha Rajanala & Ch. Nandini (2017). Mathematical Interpretation of Cancer Cells. *J Pharm Res*, 2017; 6(Suppl 1): S-21. https://jprinfo.com/storage/live/2020-04-06%2000_44_10jntu.pdf

CERTIFICATIONS AND TRAINING COURSES:

Human Research: Group 2: Biomedical Research Investigators: 2 – Refresher Course

George Mason University, Collaborative Institutional Training Initiative (CITI) Program Course, completed on September 25th, 2024.

Human Research: Group 2: Biomedical Research Investigators: 1 – Basic Course

George Mason University, Collaborative Institutional Training Initiative (CITI) Program Course, completed on September 5th, 2023.

Life Sciences and Health Care Analytics: R Programming Course

360DigiTMG Course, completed on December 18th, 2020.

Antimicrobial Stewardship: A Competency – Based Approach Course

World Health Organization Course, completed on May 25th, 2020.

The Simplest Guide to Clinical SAS Programming Course

Udemy Course, completed on May 19th, 2020.

Vital Signs: Understanding What the Body is Telling Us Course

University of Pennsylvania, Coursera Course, completed on April 24th, 2020.

ORGANIZATION MEMBERSHIPS:

HIMSS Student Member

Healthcare Information and Management Systems Society (HIMSS), National Capital Area (Area), Washington D.C.
Membership since February 2024.

Phi Kappa Phi Student Member

The Honor Society of Phi Kappa Phi, University of Maine, U.S.A.
Membership since April 2024.